

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 10%

QUESTIONS:50

Total Marks:25

Annexure A

Multiple Choice Questions (MCQ)

Code of Conduct:

I Abinesh H (192424387) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Abinesh H

Annexure A

Short Answer Test

1. Bayes' Theorem is a fundamental concept in probability theory. Which of the following best describes its primary purpose in updating prior beliefs when new evidence is observed in a probabilistic model?

- a) To compute only prior probabilities
- b) To update prior beliefs using new evidence
- c) To eliminate irrelevant features
- d) To maximize classification accuracy

Correct Option: b) To update prior beliefs using new evidence

Reason: Bayes' Theorem updates prior beliefs using new evidence.

2. In the Naïve Bayes classifier, a strong assumption is made about the relationship between features. Which statement correctly describes this assumption?

- a) Features are completely dependent on each other
- b) Features are conditionally independent given the class
- c) Features must be continuous
- d) Features are always correlated

Correct Option: b) Features are conditionally independent given the class

Reason: Naïve Bayes assumes conditional independence of features given the class.

3. Maximum Likelihood Estimation is widely used for parameter estimation in probabilistic models. Which option correctly explains the objective of MLE?

- a) Maximize prior probability
- b) Minimize hypothesis space
- c) Maximize likelihood of observed data
- d) Reduce computational complexity

Correct Option: c) Maximize likelihood of observed data

Reason: MLE chooses parameters that maximize the likelihood of observed data.

4. The Minimum Description Length principle is often used in model selection. Which best explains how MDL balances model complexity and goodness of fit?

- a) Prefers highly complex models always
- b) Prefers simplest model that explains data well
- c) Ignores data noise completely
- d) Focuses only on training accuracy

Correct Option: b) Prefers simplest model that explains data well

Reason: MDL favors a simple model that explains the data well.

5. The Bayes Optimal Classifier is considered theoretically ideal in classification tasks. Which statement best describes how it achieves the minimum possible classification error?

- a) By selecting random hypotheses
- b) By averaging predictions over all hypotheses
- c) By using only prior probabilities
- d) By minimizing dataset size

Correct Option: b) By averaging predictions over all hypotheses

Reason: It combines predictions from all hypotheses weighted by their posterior probabilities.

6. The Gibbs Algorithm is an important concept in Bayesian learning. Which option best explains how it selects a hypothesis from the hypothesis space during classification?

- a) Selects hypothesis randomly from consistent set
- b) Selects hypothesis with minimum loss only
- c) Selects fixed predefined hypothesis
- d) Ignores hypothesis space

Correct Option: a) Selects hypothesis randomly from consistent set

Reason: Gibbs randomly selects a hypothesis from the consistent set according to the learning framework.

7. Naïve Bayes classifiers are often used in text classification tasks. Which condition ensures that the classifier performs effectively despite its simplifying assumptions?

- a) Strong dependence between features
- b) Large amount of labeled data
- c) Features being conditionally independent
- d) Absence of noise in dataset

Correct Option: c) Features being conditionally independent

Reason: Conditional independence matches the key Naïve Bayes assumption.

8. Bayesian Belief Networks are graphical models that represent probabilistic relationships. Which best describes the role of nodes and edges?

- a) Nodes represent variables and edges represent dependencies
- b) Nodes represent datasets and edges represent features
- c) Nodes represent outputs only
- d) Edges represent noise in data

Correct Option: a) Nodes represent variables and edges represent dependencies

Reason: Nodes represent variables, while edges show probabilistic dependencies.

9. The Expectation-Maximization algorithm is commonly used when dealing with incomplete data. Which best explains the role of the Expectation step?

- a) Updates model parameters
- b) Computes expected values of hidden variables
- c) Removes missing data
- d) Sorts the dataset

Correct Option: b) Computes expected values of hidden variables

Reason: The E-step estimates expected values of hidden or latent variables.

10. In the EM algorithm, the Maximization step updates certain parameters. Which correctly describes what is updated?

- a) Hidden variables only
- b) Observed data values
- c) Model parameters maximizing likelihood
- d) Training dataset size

Correct Option: c) Model parameters maximizing likelihood

Reason: The M-step updates parameters to maximize the expected likelihood.

11. Probability learning involves estimating probability distributions from data. Which best explains its goal?

- a) To remove all uncertainty
- b) To estimate underlying probability distributions
- c) To reduce number of features
- d) To simplify model structure

Correct Option: b) To estimate underlying probability distributions

Reason: Probability learning estimates the distributions underlying the data.

12. Sample complexity is an important concept in learning theory. Which best describes what it indicates?

- a) Amount of computation required
- b) Number of samples needed for learning
- c) Model accuracy only
- d) Memory consumption

Correct Option: b) Number of samples needed for learning

Reason: Sample complexity measures how many examples are needed for effective learning.

13. A finite hypothesis space contains a limited number of possible hypotheses. Which best explains its impact?

- a) Improves flexibility always
- b) Restricts possible models considered
- c) Eliminates need for data
- d) Guarantees perfect accuracy

Correct Option: b) Restricts possible models considered

Reason: A finite space limits the models that can be considered.

14. In contrast to finite hypothesis spaces, infinite hypothesis spaces allow more flexibility. Which best describes their characteristics?

- a) Limited to discrete functions
- b) Allows continuous and complex models
- c) Contains no hypotheses
- d) Simplifies computation always

Correct Option: b) Allows continuous and complex models

Reason: Infinite spaces can contain continuous and highly complex models.

15. The mistake bound model is used to evaluate learning algorithms. Which best describes what is measured?

- a) Training time
- b) Maximum number of mistakes made
- c) Memory usage
- d) Dataset size

Correct Option: b) Maximum number of mistakes made

Reason: The mistake bound measures the maximum prediction mistakes in online learning.

16. Prior probability is a key concept in Bayesian inference. Which best describes what prior probability represents?

- a) Probability after evidence
- b) Initial belief before observing data
- c) Conditional probability only
- d) Joint probability

Correct Option: b) Initial belief before observing data

Reason: A prior is the belief assigned before new evidence is observed.

17. Posterior probability is calculated after incorporating evidence. How is it derived using Bayes' Theorem?

- a) Using only likelihood
- b) Using prior and likelihood with evidence
- c) Ignoring prior information
- d) Using random sampling

Correct Option: b) Using prior and likelihood with evidence

Reason: Posterior probability combines the prior, likelihood, and evidence.

18. Likelihood plays a central role in statistical inference. What does likelihood represent?

- a) Probability of hypothesis given data
- b) Probability of data given hypothesis
- c) Probability of prior only
- d) Probability of error

Correct Option: b) Probability of data given hypothesis

Reason: Likelihood measures how probable the observed data is under a hypothesis.

19. Naïve Bayes classifiers are widely used in real-world applications. Which type of problems is it most effective for?

- a) Clustering problems
- b) Classification problems with high-dimensional data
- c) Regression problems
- d) Sorting problems

Correct Option: b) Classification problems with high-dimensional data

Reason: Naïve Bayes is efficient for high-dimensional classification tasks.

20. Overfitting is a common issue in machine learning. Which situation describes overfitting?

- a) Model performs well on unseen data
- b) Model captures noise instead of pattern
- c) Model is too simple
- d) Model ignores data

Correct Option: b) Model captures noise instead of pattern

Reason: Overfitting occurs when a model learns noise rather than general patterns.

21. Underfitting occurs when a model fails to capture patterns in data. Which best describes it?

- a) High accuracy always
- b) Fails to learn patterns from data
- c) Too complex model

d) Overtrained model

Correct Option: b) Fails to learn patterns from data

Reason: Underfitting means the model is too simple to learn important patterns.

22. The EM algorithm is iterative and requires a stopping condition. When is it considered to have converged?

- a) When data changes
- b) When parameters stabilize across iterations
- c) When iterations increase
- d) When accuracy decreases

Correct Option: b) When parameters stabilize across iterations

Reason: Convergence occurs when parameter updates become sufficiently small or stable.

23. Bayesian learning combines prior knowledge with observed data. How do these components interact?

- a) Prior replaces data
- b) Data replaces prior
- c) Both are combined to update belief
- d) Neither is used

Correct Option: c) Both are combined to update belief

Reason: Bayesian learning combines prior information with observed evidence.

24. Concept learning involves identifying a suitable hypothesis. What is its main goal?

- a) To reduce computation
- b) To find hypothesis consistent with data
- c) To eliminate features
- d) To increase dataset size

Correct Option: b) To find hypothesis consistent with data

Reason: Concept learning searches for a hypothesis consistent with the training examples.

25. Hypothesis space search is a key component of concept learning. How does it help select an appropriate model?

- a) Random selection of hypothesis
- b) Exploring possible hypotheses to find best fit
- c) Ignoring data patterns
- d) Reducing dataset size

Correct Option: b) Exploring possible hypotheses to find best fit

Reason: Searching the hypothesis space helps find a model that best fits the data.

26. In Bayesian inference, normalization ensures probabilities sum to one. What role does the evidence term play in Bayes' Theorem?

- a) It acts as a scaling factor
- b) It removes prior probability
- c) It increases model complexity
- d) It selects hypothesis randomly

Correct Option: a) It acts as a scaling factor

Reason: The evidence term normalizes the posterior probabilities.

27. In Naïve Bayes classification, what is the primary reason for using logarithms when computing probabilities?

- a) To increase accuracy
- b) To avoid numerical underflow
- c) To remove dependencies
- d) To simplify dataset

Correct Option: b) To avoid numerical underflow

Reason: Logarithms convert tiny probability products into manageable sums.

28. Which of the following best describes conditional probability?

- a) Probability of two events occurring together
- b) Probability of an event given another event has occurred
- c) Probability of independent events
- d) Probability of random variables

Correct Option: b) Probability of an event given another event has occurred

Reason: Conditional probability measures an event given another event.

29. Which factor most affects the performance of a Naïve Bayes classifier in real-world datasets?

- a) Number of classes
- b) Independence assumption violation
- c) Dataset size only
- d) Output format

Correct Option: b) Independence assumption violation

Reason: Strong dependence among features violates the model's key assumption.

30. In Bayesian networks, what ensures that the model remains computationally efficient?

- a) Large dataset
- b) Conditional independence
- c) Continuous variables
- d) High dimensionality

Correct Option: b) Conditional independence

Reason: Conditional independence reduces the number of probabilities that must be modeled.

31. Which best describes the role of prior knowledge in Bayesian learning?

- a) Eliminates need for data
- b) Guides learning before observing data
- c) Reduces accuracy
- d) Replaces hypothesis

Correct Option: b) Guides learning before observing data

Reason: Prior knowledge provides an initial belief before new data is observed.

32. In Maximum Likelihood Estimation, what happens when the dataset size increases significantly?

- a) Prior dominates
- b) Likelihood estimation becomes more accurate
- c) Model becomes random
- d) Complexity decreases

Correct Option: b) Likelihood estimation becomes more accurate

Reason: More data generally gives a more reliable likelihood estimate.

33. Which of the following is a limitation of MLE?

- a) Uses too much prior knowledge
- b) Ignores prior information
- c) Cannot handle data
- d) Always overfits

Correct Option: b) Ignores prior information

Reason: MLE uses the likelihood and does not incorporate prior information.

34. Which best describes Bayesian estimation compared to MLE?

- a) Uses only likelihood
- b) Uses both prior and likelihood
- c) Ignores uncertainty
- d) Works only for small datasets

Correct Option: b) Uses both prior and likelihood

Reason: Bayesian estimation combines prior information with the likelihood.

35. What is the main advantage of Laplace smoothing in Naïve Bayes?

- a) Removes features
- b) Handles zero probability problem
- c) Reduces dataset size
- d) Increases bias only

Correct Option: b) Handles zero probability problem

Reason: Laplace smoothing prevents unseen feature values from producing zero probability.

36. Which best defines a hypothesis in machine learning?

- a) Final output
- b) Assumed model for mapping inputs to outputs
- c) Dataset
- d) Feature vector

Correct Option: b) Assumed model for mapping inputs to outputs

Reason: A hypothesis is a candidate model mapping inputs to predicted outputs.

37. What is the purpose of hypothesis space in learning algorithms?

- a) Store data
- b) Define all possible models
- c) Reduce training time
- d) Increase features

Correct Option: b) Define all possible models

Reason: The hypothesis space contains all candidate models considered by the learner.

38. Which best explains generalization in machine learning?

- a) Memorizing training data
- b) Performing well on unseen data
- c) Reducing dataset size
- d) Increasing parameters

Correct Option: b) Performing well on unseen data

Reason: Generalization is the ability to perform well on new, unseen data.

39. What is the effect of noisy data on learning algorithms?

- a) Improves accuracy
- b) Leads to overfitting
- c) Eliminates features
- d) Reduces complexity

Correct Option: b) Leads to overfitting

Reason: Noise can be memorized by the model, increasing overfitting.

40. In EM algorithm, what is the main challenge during initialization?

- a) Dataset size
- b) Choosing initial parameters
- c) Output format
- d) Removing features

Correct Option: b) Choosing initial parameters

Reason: EM can depend on the choice of initial parameter values.

41. Which best describes convergence in iterative algorithms?

- a) Increasing error
- b) Stabilization of parameters
- c) Random updates
- d) Dataset reduction

Correct Option: b) Stabilization of parameters

Reason: Convergence means the parameter values become stable.

42. Which is a characteristic of probabilistic models?

- a) Deterministic outputs
- b) Handling uncertainty
- c) Ignoring data
- d) Fixed predictions

Correct Option: b) Handling uncertainty

Reason: Probabilistic models explicitly represent uncertainty.

43. What is the role of likelihood in Bayesian inference?

- a) Represents prior
- b) Represents evidence
- c) Measures fit of data to hypothesis
- d) Removes uncertainty

Correct Option: c) Measures fit of data to hypothesis

Reason: Likelihood measures how well the observed data fits a hypothesis.

44. Which best describes overfitting?

- a) Model too simple
- b) Model fits noise in training data
- c) Model ignores data
- d) Model has low variance

Correct Option: b) Model fits noise in training data

Reason: Overfitting occurs when the model learns training noise.

45. Which best describes underfitting?

- a) Model too complex
- b) Model fails to capture patterns
- c) Model memorizes data
- d) Model has high variance

Correct Option: b) Model fails to capture patterns

Reason: Underfitting occurs when the model cannot capture important patterns.

46. What is the role of training data in learning algorithms?

- a) Defines output only
- b) Guides model learning
- c) Reduces complexity
- d) Eliminates hypothesis

Correct Option: b) Guides model learning

Reason: Training data provides examples from which the model learns.

47. Which factor improves model generalization?

- a) Increasing noise
- b) Proper regularization
- c) Reducing data
- d) Ignoring validation

Correct Option: b) Proper regularization

Reason: Regularization helps control model complexity and reduce overfitting.

48. Which best explains feature selection?

- a) Removing dataset
- b) Selecting relevant features
- c) Increasing features
- d) Ignoring attributes

Correct Option: b) Selecting relevant features

Reason: Feature selection keeps the most relevant input variables.

49. Which best describes classification problems?

- a) Predict continuous values
- b) Assign labels to data
- c) Cluster data
- d) Reduce dimensions

Correct Option: b) Assign labels to data

Reason: Classification assigns data instances to discrete classes or labels.

50. Which best describes regression problems?

- a) Predict discrete labels
- b) Predict continuous values
- c) Cluster data
- d) Classify data

Correct Option: b) Predict continuous values

Reason: Regression predicts continuous numerical values.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 - Exemplary (5 Marks)	Level 3 - Proficient (4 Marks)	Level 2 - Developing (2-3 Marks)	Level 1 - Beginning (0-1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand